

LE NGUYEN DUY LUAN

Android & AI Engineer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Android & AI Engineer specializing in Kotlin, Jetpack Compose, and on-device AI applications. Built end-to-end Android applications using Clean Architecture, MVVM, CameraX, and TensorFlow Lite, with a focus on offline-first experiences and efficient on-device inference.

EDUCATION

Nha Trang University

Bachelor of Information Technology

• **GPA:** 3.25/4.0

2022 – 2026

Nha Trang, Vietnam

TECHNICAL SKILLS

Architecture: Clean Architecture, MVVM, Repository Pattern, StateFlow, Coroutines

Android: Jetpack Compose, CameraX, Room, WorkManager, Navigation, Paging3, DataStore

AI & CV: TensorFlow Lite, MediaPipe, ML Kit, OpenCV, YOLO (Detection & Classification)

Backend & Cloud: Firebase, Supabase, REST API, Retrofit, OkHttp

Programming Languages: Kotlin, Java, Python, SQL, JavaScript

Tools: Git, Android Studio, Gradle, ADB, KSP, VS Code

Languages: Vietnamese (Native) – English (Intermediate, comfortable reading technical documentation)

PROJECTS

[VocabLensAI](#) — Native Android English Learning Platform | *Personal Project*

- **Tech Stack:** Kotlin, Jetpack Compose, Coroutines/Flow, Room DB, CameraX, YOLO11m (TFLite, INT8), Gemini API, Firebase, Supabase.
- Designed and developed an offline-first Android application using Kotlin, Jetpack Compose, MVVM, and Clean Architecture, supporting AI-powered vocabulary learning with real-time object recognition.
- Implemented real-time on-device object detection using an INT8-quantized YOLO11m model deployed with TensorFlow Lite and CameraX, while integrating the Gemini API as an optional cloud service for contextual vocabulary story generation.
- Implemented the SM-2 spaced repetition algorithm to optimize vocabulary review scheduling, backed by a hybrid Room DB/Firestore synchronization mechanism for reliable offline-first data consistency.
- Built a real-time multiplayer PvP Arena using Firebase Firestore with an ELO rating calculator, alongside a chess-style game review UI for match accuracy analysis and AI feedback.

[Lung X-ray Trustworthy Classifier \(Android + Research\)](#) | *Graduation Project, NTU* (Final Grade: 9.1/10)

- **Tech Stack:** Python, TFLite, U-Net (ResNet50), YOLOv12m-cls, Kotlin, Jetpack Compose, Groq API.
- Deployed a 2-stage AI pipeline (U-Net segmentation → YOLOv12m-cls image classification) to a native Android app via TFLite, enabling fully offline on-device inference across 4 pathology classes (COVID-19, Viral Pneumonia, Lung Opacity, Normal).
- Improved model reliability by engineering median-fill masking to remove non-lung visual artifacts before classification, improving visual localization and model interpretability. The model was trained on nearly 12,000 chest X-ray images and evaluated on 300 expert-annotated localization cases.
- Built an OOD safety gate (lung-area ratio thresholding, 8–72% valid range) to auto-reject invalid inputs, and integrated dual-mode XAI (Occlusion Sensitivity + Grad-CAM++).
- Added a context-aware LLM assistant (Groq/LLaMA 3.1, Fleischner-guideline grounded) and native PDF report export, benchmarked against 5 backbone architectures.

[AI Fitness Coach](#) | *Personal Project*

- **Tech Stack:** Kotlin, Jetpack Compose, CameraX, MediaPipe/MLKit (Pose Estimation), Room DB, Text-To-Speech (TTS).
- Developed an offline-first Android fitness application featuring an 800+ exercise library and a personalized workout recommendation engine built with Kotlin, Jetpack Compose, and Room.
- Addressed the injury risk of unsupervised home workouts by implementing camera-based pose estimation (MediaPipe) to extract 33 keypoints and calculate 3D vector angles in real-time.
- Built a custom rule-based feedback engine with Text-To-Speech integration (audio alerts) and automated rep counting, caching progress 100% offline via Room DB.